ROCKS

1. Rocks and Minerals 🪨💎

Definitions:

- **Rock:** An aggregate of minerals that forms a more or less definite unit of the earth's crust. It refers to hard solid matter like granite, and also to soft/loose particles like sand, silt, and clay derived from the earth.
- **Minerals:** Naturally occurring inorganic solids with a definite chemical composition and a crystalline structure.
- **Element:** A pure substance.

Key Points:

- Rocks and minerals are closely related but differ from one another; rocks are composed of minerals, but minerals are not composed of rocks.
- The earth's crust is made up of various types of rocks that differ in texture, structure, colour, permeability, mode of occurrence, and degree of resistance to denudation.
- Elements are organised to form compounds which are known as minerals.

The "Why?" Section ❓

- **Why is a rock considered "impure" compared to an element?** Because an element is a pure substance, whereas a rock has no definite chemical composition and is an aggregate of multiple minerals.

Things to Remember 🧠

- **Most abundant mineral elements (Earth as a whole):** Iron, silicon, magnesium, nickel, sulphur, and calcium.
- **Most abundant mineral elements (Earth's crust):** Silicon, aluminium, iron, magnesium, calcium, potassium, and sodium.
- **Four chief mineral groups:** Silicates, carbonates, sulphides, and metallic minerals.
- **Important Rock Examples:** Basalt, granite, sandstone, slate, and quartz.

Table of Differences 🆚

Feature	Rocks 🪨	Minerals 💎
Basic Nature	Aggregates of minerals.	Solid inorganic substances occurring naturally.
Chemical Composition	No definite chemical composition (impure).	Have a definite chemical composition.
Organization	Made of organised minerals.	Made of organised elements forming compounds (pure).

2. Types of Rocks & Igneous Rocks 🔥🌋

Definitions:

- **Igneous Rocks:** Rocks formed due to the cooling, solidification, and crystallisation of hot molten material (magma) found at great depths in the interior of the earth. The word is derived from the Latin word 'ignis' meaning fire.
- **Sialic Rocks:** Igneous rocks composed of alumino-silicate minerals (Silica + Aluminium) like quartz and feldspar to granite and rhyolite. They are light and buoyant.
- **Primary Rocks:** Another name for igneous rocks because they were the first to be formed and form the basis of formation for all other rock types.

Key Points:

- Rocks are classified in several ways, but the most important classification is done on the basis of their origin into three main types: Igneous, Sedimentary, and Metamorphic.
- Igneous rocks are hard, compact, granular, and crystalline.
- They form by the solidification of molten magma.

- Their texture can be fine-grained/smooth or coarse with large crystals, strictly depending upon the rate of cooling of the molten material.
- Most consist of silicates, do not have layers, and are generally weathered by mechanical weathering rather than chemical weathering.
- They are associated with volcanic activity and found mostly in volcanic regions.

The "Why?" Section ❓

- **Why are igneous rocks less affected by chemical weathering?** Because they are hard, compact, and non-porous, meaning water does not percolate into these rocks.

3. Classification of Igneous Rocks (Based on Origin) 🏔️🔽

Definitions:

- **Extrusive Igneous Rocks (Volcanic Rocks):** Formed by the cooling of molten magma that is brought to the earth's surface through fissures or volcanic eruptions.
- **Intrusive Igneous Rocks:** Formed when magma solidifies slowly below the earth's surface, forming coarse-textured, hard rocks with large crystals.
- **Plutonic Rocks:** Deep-seated intrusive igneous rocks that form large crystals due to the very slow cooling of magma at great depth.
- **Hypabyssal Rocks:** Minor intrusive rocks formed when magma cools at intermediate depths (e.g., sills and dykes).

Key Points (Intrusive Landforms & Examples):

- **Extrusive Rocks:** Due to *instant solidification* on the surface, they form small crystals and are smooth, crystalline, and fine-grained. Basalt is a common example, forming lava flows, sheets, and plateaus. Some basalt forms long polygonal columns, like the Giant's Causeway in Northern Ireland, which comes into view due to denudation.
- **Intrusive Rocks:** They take a *long time to cool down* inside the crust, leading to large crystal formation (e.g., granite, gabro, dolerite).

- **Batholiths:** Deep-seated, dome-shaped intrusions with no definite floor forming the cores of mountain ranges ('Bathos' = depth). They are exposed to the surface through continued erosion precisely because they are intrusive rocks (e.g., Ranchi Batholiths).
- **Laccoliths:** Formed just below the crust when magma cannot reach the surface; they have a dome-shaped upper surface and a flat bottom (Greek 'lakkos' = cistern).
- **Sills:** Formed when magma flows horizontally between rock layers and hardens.
- **Dykes:** Formed when magma is forced upwards, filling vertical cracks or fissures in existing rocks.
- **Necks:** Formed when the passage of an extinct volcano is filled with magma that solidifies into a plug.

4. Classification of Igneous Rocks (Based on Chemical Composition) 🧪

Definitions:

- **Acid Igneous Rocks:** Igneous rocks with high silica content (65% to 85%), generally lacking in iron and magnesium.
- **Basic Igneous Rocks:** Igneous rocks with lower silica content (40% to 60%) and a higher percentage of oxides of denser elements.

Key Points:

- Igneous rocks are usually compact, of interest to builders and sculptors, and may be associated with metal ores.
- Important examples across both types include gabro, basalt, granite, dolerite, feldspar, pumice, obsidian, permadite, and soria.

Table of Differences 🆚

Feature	**Acid Igneous Rocks** ⚪	**Basic Igneous Rocks** ⚫
Silica Content	High (65% to 85%).	Lower (40% to 60%).

Feature	Acid Igneous Rocks ⚪	Basic Igneous Rocks ⚫
Composition	Lacks iron and magnesium.	Higher percentage of oxides of denser elements.
Appearance & Density	Lighter in color, composed of large crystals (quartz, feldspar, mica).	Heavy and dark in colour.
Examples	Granite (commonly used as building material).	Basalt, dolerite, and gabro.

5. Sedimentary Rocks (Secondary Rocks) 🌊🪨

Definitions:

- **Sedimentary Rocks:** Secondary rocks formed by the solidification of sediments derived from original igneous, sedimentary, or metamorphic parent rocks. Examples include sandstone and shale.
- **Lithification:** The core process that turns loose sediments into hard, compacted rock (lith).
- **Clastic Sediments:** Fragments resulting from the breaking up of other older rocks through denudation by water, glaciers, and wind.
- **Stratified Rocks:** Rocks formed in horizontal layers (strata) due to the gradual squeezing of overlapping sediments.

Key Points:

- When igneous rocks are exposed on the surface, they weather and are carried away by rivers, glaciers, and ocean waves.
- These sediments deposit over long periods in seas, lakes, and streams.
- Subsequent layers are added, and their own downward weight cements them together in a process known as lithification.

- They are the most widespread rocks, constituting about 75% of the earth's surface area.
- They are generally not crystalline, are soft compared to igneous rocks, and are layered.
- They make extensive landforms but do not make massive interior forms like batholiths.

The "Why?" Section ❓

- **Why do sedimentary rocks contain fossils?** Because they are formed by surface materials derived from other rocks and environments, which means plant and animal remains frequently get trapped in the accumulating layers.

Things to Remember (The 3 Lithification Processes) 🧠

- **(i) Evaporation:** Water from loose materials is evaporated, and the remaining material accumulates before compaction (e.g., rock-salt).
- **(ii) Compaction:** Piled-up sediments are gradually squeezed by the sheer weight of the overlying layers, turning materials like sand into sandstone.
- **(iii) Cementation:** The bonding together of compacted sediments by natural cementing elements/compounds like calcium, silica, and iron.

Table of Differences (Classification based on Formation) 🆚

Formation Type	**Mechanically Formed 🔨**	**Chemically Formed 🧪**
Process	Breaking up of older rocks via denudation, followed by lithification (squeezing of layers).	Formed by direct precipitation of mineral matter from a solution. When part is dissolved, some part precipitates down.

Formation Type	Mechanically Formed 🔨	Chemically Formed 🧪
Rock Characteristics	Also known as Clastic or Stratified rocks.	Formed by chemical accumulation.
Examples	Shale, sandstone, clay, conglomerates, and loess.	*(Text cutoff in source, but involves precipitated minerals).*

5. Sedimentary Rocks (Classification on the Basis of Formation) 🌊🪨

(Repaired Section)

Table of Differences 🆚

Formation Type	Mechanically Formed 🔨	Chemically Formed 🧪
Process	Formed from fragments resulting from the breaking up of other older rocks (Clastic Sediments) through denudation by agents like water, moving glaciers, and wind.	Formed by the direct precipitation of mineral matter from solution. When part is dissolved, some part precipitates down.
Action	Sediments are gradually squeezed by the weight of overlying layers to form strata (Stratified rocks).	Accumulation takes place in **lakes and lagoons**. They are compacted strictly through **evaporation**.
Examples &	Shale, sandstone, clay, conglomerates, and loess.	**Gypsum, rock salt, and potash**. Gypsum is found globally from evaporated

Formation Type	Mechanically Formed 🔨	Chemically Formed 🧪
Locations 📌		seawater. Rock salt is found in the **Dead Sea, Aral Sea, and Sambhar Lake in Rajasthan!**

6. Organically Formed Sedimentary Rocks 🌿🪨

Definitions:

- **Organically Formed Rocks:** Rocks containing the remains of dead plants and animals. 🐾
- **Carbonaceous Rocks:** Rocks formed from vegetation/plant matter, such as peat, lignite, bituminous, and anthracite. 🪵
- **Siliceous Rocks:** Rocks formed due to the dominance of silica contents. 💎
- **Fossil Fuels:** The remains of animals and plants which have become hard and turned into sedimentary rocks/fuels such as coal or petroleum. ⛽

Key Points (Exhaustive Detail):

- Limestone (or calcareous rocks) is formed strictly by skeletons, shells, and animal remains. It contains large proportions of lime. 🐚
- Chalk is a specific form of carbonate rock. It is formed directly due to the precipitation of carbonate materials. ⚪
- In the ancient past, there were vast swamps of forest regions that got buried and underwent changes to yield fossil fuels. 🌲
- The vegetable matter undergoes changes after being submerged in underground water. 💧
- Formation into rocks proceeds in distinct stages: peat -> lignite -> coal. 🔄
- The products of the first phase are *peat*, which is used as fuel. 🔥

- *Lignite* is a more decomposed rock of organic matter and is also used as fuel. 🪨
- Several new products have been obtained from these rocks in recent years, including wax and resins. 🕯️
- In coal, the percentage of carbon is very high. Two types of coal, *bituminous* and *anthracite*, are massive sources of fuel for power generation. ⚡ ⬛

The "Why?" Section ❓

- **Why is fossilisation so rare?** Fossilisation requires the animal to be perfectly buried and the deposition of material over the site. Because land is mostly eroded, fossilisation is rare. Deposition on land occurs only in river valleys. In the open sea, the strong currents simply break and wear down the animal remains. Finally, for animals without skeletons, fossilisation is rare as there are no bones! 🦴

Things to Remember (The 6 Steps of Petrification/Fossil Formation) 🧠

- 1. The animal dies. ☠️
- 2. Scavengers consume them. 🦅
- 3. Soft parts (tissues, muscles, organs, etc.) start to decompose. 🦠
- 4. Mud, sand, and silt strictly cover the body (usually only bones and teeth remain). 🏖️
- 5. More layers of sediment build-up, putting massive pressure on the layers below, eventually leading to the formation of sedimentary rocks. 🗜️

6. During the build-up, water seeps into the bones and teeth, leaving behind mineral crystals. ✨

7. Sedimentary Rocks (Classification on the Basis of Agents of Formation) 🌊🌬️

Definitions:

- **Riverine Rocks:** Formed by the alluvial deposits brought by the flowing water of streams. 🏞️
- **Lacustrine Rocks:** Found exactly on the bed of a lake corresponding to successive periods of deposition. 💧
- **Glacial Rocks:** Formed by glacial deposits in the form of debris or tills (like boulders, gravels, sand, etc.). 🧊
- **Aeolian Rocks:** Rocks formed with sand particles brought by winds. The deposition makes them hard (found in deserts). 🐪
- **Marine Rocks:** Formed deep at the ocean floor. ⚓

Key Points:

- Glaciers heavily erode the surface and the sides of a valley and transport the eroded material further. ⛏️
- When the glacier melts due to heat, the debris brought by it is left behind in the form of moraines, which form glacial rocks. 🌰
- Loess is an important Aeolian deposit. 🌬️

Table of Differences (Types of Marine Rocks) 🆚

Feature	**Calcarious Sedimentary Marine Rocks** 🐚	**Carbonaceous Sedimentary Marine Rocks** 🌱
Formation Source	Deposition of shells and skeletons of sea organisms (corals, clams, oysters).	Sea plants which remain buried for a very long period.

Feature	Calcarious Sedimentary Marine Rocks 🐚	Carbonaceous Sedimentary Marine Rocks 🌱
Mechanism	Organisms extract calcium carbonate from ocean water.	Long-preserved remains are converted into coal, lignite, and peat due to the extreme pressure of overlying rocks.
Examples	Chalk and limestone.	Coal, lignite, and peat.

8. Metamorphic Rocks 💎🔥

Definitions:

- **Metamorphism:** Derived from 'metamorphose' which strictly means "change in form". It refers to the massive alteration of the composition or structure of a rock by heat, pressure or other natural agency. 🔄

Key Points:

- These rocks were once completely normal igneous or sedimentary rocks which underwent change through severe physical or chemical processes. 🪨➡️💎
- The factors that take part in changing the form of rocks are strictly temperature and pressure, or both. 🌡️🗜️
- Since rocks are composed of minerals, these may absolutely change their composition and texture under great pressure and temperature. ✨
- **Characteristic 1:** They are significantly harder and more compact than their original form (e.g., marble made from limestone is harder than limestone). 🧱
- **Characteristic 2:** Most of them are totally impermeable — they do not allow water to percolate through them. 🛡️
- **Characteristic 3:** They absolutely do not have fossils. 🚫🦴
- **Characteristic 4:** Metamorphic rocks are formed strictly due to change in the texture and composition of the pre-existing rocks. 🧩

- **Characteristic 5:** Entirely new minerals are formed during the intense process of metamorphism. 🔮

The "Why?" Section ❓

- **Why do metamorphic rocks never contain fossils?** Because any fossils that may have existed in the original sedimentary rocks are completely destroyed during the extreme heat and crushing pressure of the formation of metamorphic rocks! 🔥💥

9. Types of Metamorphism 🌋🧲

Definitions:

- **Thermal Metamorphism:** Occurs when the transformation of the original rock takes place strictly due to the influence of high temperature. 🔥
- **Dynamic Metamorphism:** Occurs when transformation takes place mainly because of massive pressure at a great depth within the earth's crust. 🧲
- **Foliation:** When minerals get arranged in a series of bands under conditions of dynamic change. 〰️
- **Regional Metamorphism:** Occurs due to the mountain building process where igneous/sedimentary rocks are buried deep inside the crust, altered by the pressure of overlying rocks and intense heat from large-scale earth movements. 🏔️
- **Local or Contact Metamorphism:** Occurs when a small area is affected by hot magma affecting cracks or layers of rocks, transforming them. 🎈

Key Points:

- In Thermal Metamorphism, the heat may be directly from hot magma or from the intense friction of moving rock layers. 🌋
- For example, slate is formed from clay, and graphite is formed from coal through thermal metamorphism. 🖍️
- In Dynamic Metamorphism, chemically active hot gases passing through the rocks actively change their chemical composition. 🧪

- The rocks used in the Taj Mahal at Agra are brilliant marbles metamorphosed directly from Dolomites. 🕌
- The Dilwara Temple at Mount Abu is also built of these similar marbles. 🛕
- **Magma Intrusion Note (Handwritten Extraction!):** When a plume of hot magma rises and spreads horizontally beneath layers, it directly causes the surrounding rock to **"change its form also"** through contact metamorphism! 🌋 ➡️ 💎

Things to Remember (Original Rock to Metamorphic Transformation) 🧠

Original Rock 🪨	**Undergoes Changes to Form (Metamorphic Rock)** 💎
Limestone	Marble
Sandstone	Quartzite
Shale	Slate
Coal	Graphite
Basalt	Schist
Granite	Gneiss
Dolomite / Chalk	Marble *(Handwritten arrow noted!)*

10. Economic Significance & The Rock Cycle 🔁🌍

Definitions:

- **Rock Cycle:** This continuous process of transformation of old rocks into new ones. ♻️

Key Points (Economic Significance):

- Many metallic minerals such as gold and silver are found directly in metamorphic rocks. 🥇
- Like minerals, rocks are of great resource value, some directly and some as constituents of minerals. 💰
- Soils are derived primarily from the weathering of rocks. 🌱
- Almost all types of building materials used for paving roads, floors, or building walls/houses/bridges come straight from rocks. 🧱
- Rocks are a major source of precious metals like gold, silver, platinum, etc. ✨
- Apart from minerals and fossil fuels, even rock wastes have been actively used in manufacturing various articles. 🏭

Key Points (The Rock Cycle Process):

- The earth is said to be roughly 4.5 billion years old, and obviously, rocks undergo cyclic transformation constantly. ⏳
- The cycle strictly passes through the process of formation of sedimentary, metamorphic, and igneous rocks. 🔄
- **Step 1:** Igneous rocks are the primary source of all other rocks, formed on the outer surface due to volcanic activities. They are disintegrated/eroded by climatic factors, removed, and transported to lower levels or the sea by running water, winds, and glaciers, and ocean currents. 🌋➡️🌊
- **Step 2:** Sediments accumulate in basins, lakes, or ocean beds. The accumulations undergo lithification strictly through the processes of evaporation, compaction, and cementation as discussed earlier. 🗜️
- **Step 3:** Once loose sediment transforms into sedimentary rock, it is carried in depressions or on the sea floor further down under the earth. It may undergo further disintegration and be recrystallised into igneous rock, or transformed into metamorphic rock due to intense heat and pressure. 🔥

- **Step 4:** In case the rock melts, it may again result in the formation of igneous rock. The disintegrated material may again form sedimentary rock. 🌋
- Completing any of these extreme changes involves hundreds and thousands of years. 🕰️
- In the operation of the rock cycle, the earth is kept young and movements of adjustments, especially through horizontal flow or convection currents, maintain the vital balance between various elements of the earth like oceans and continents! ⚖️ 🌍